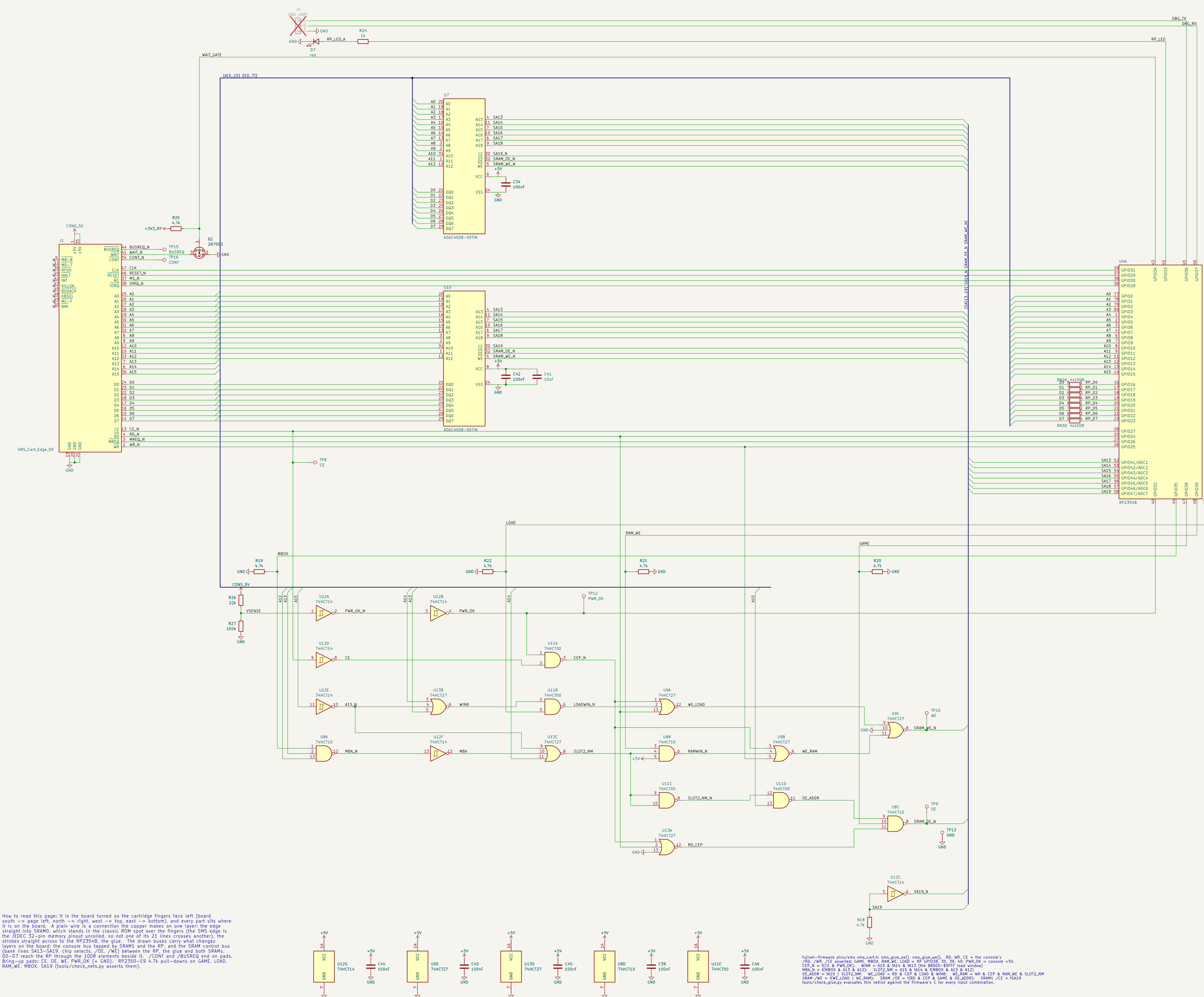
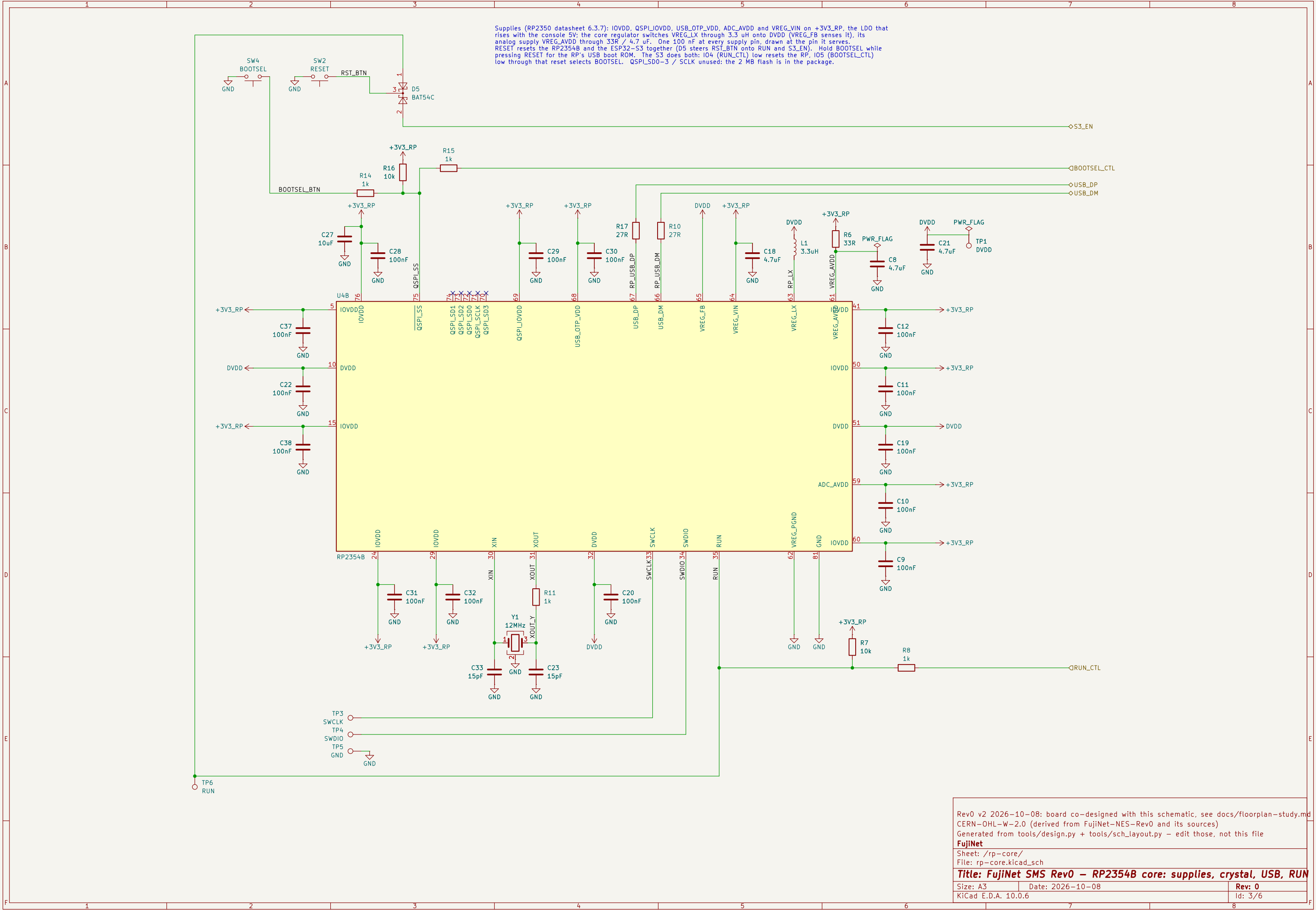
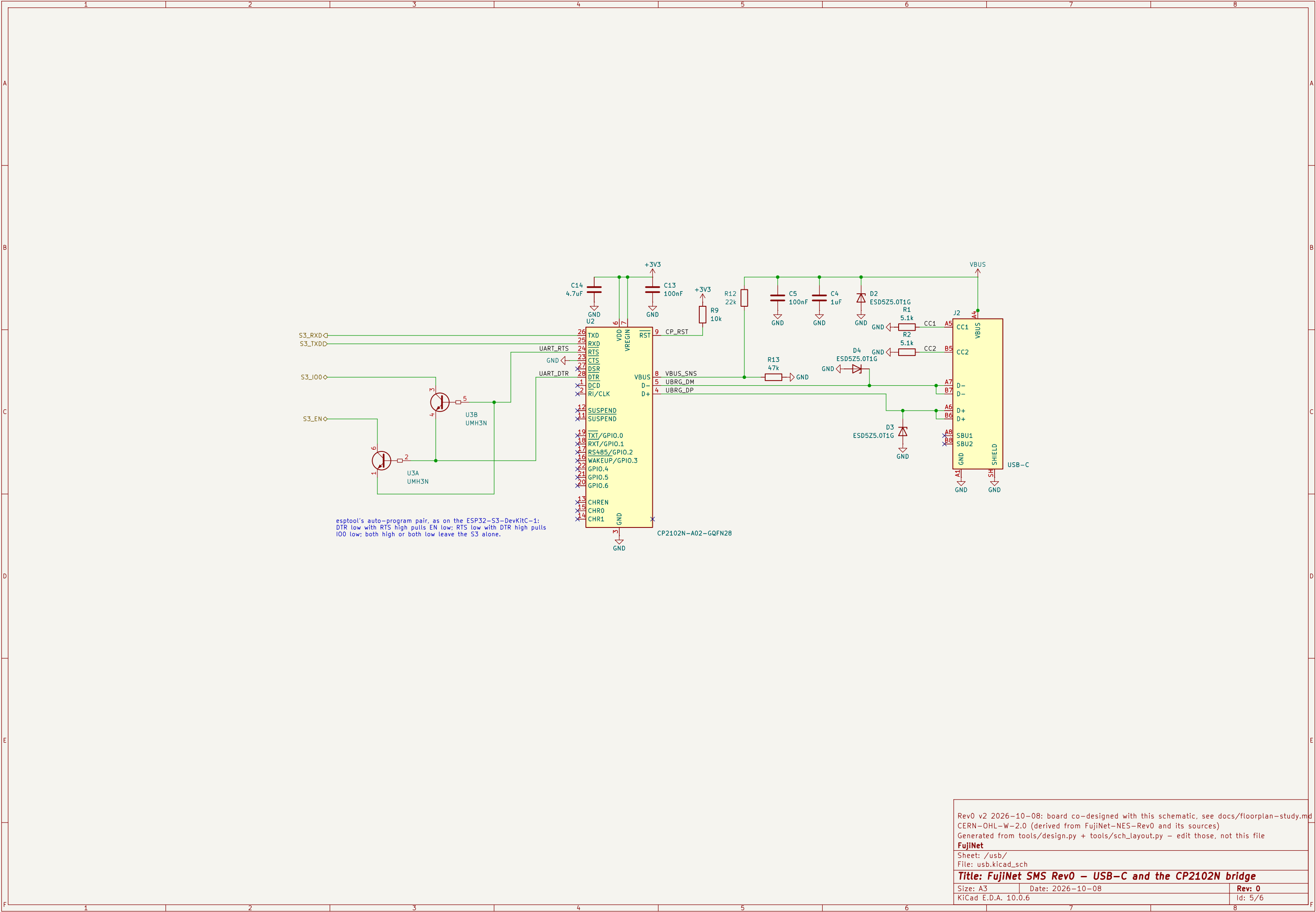




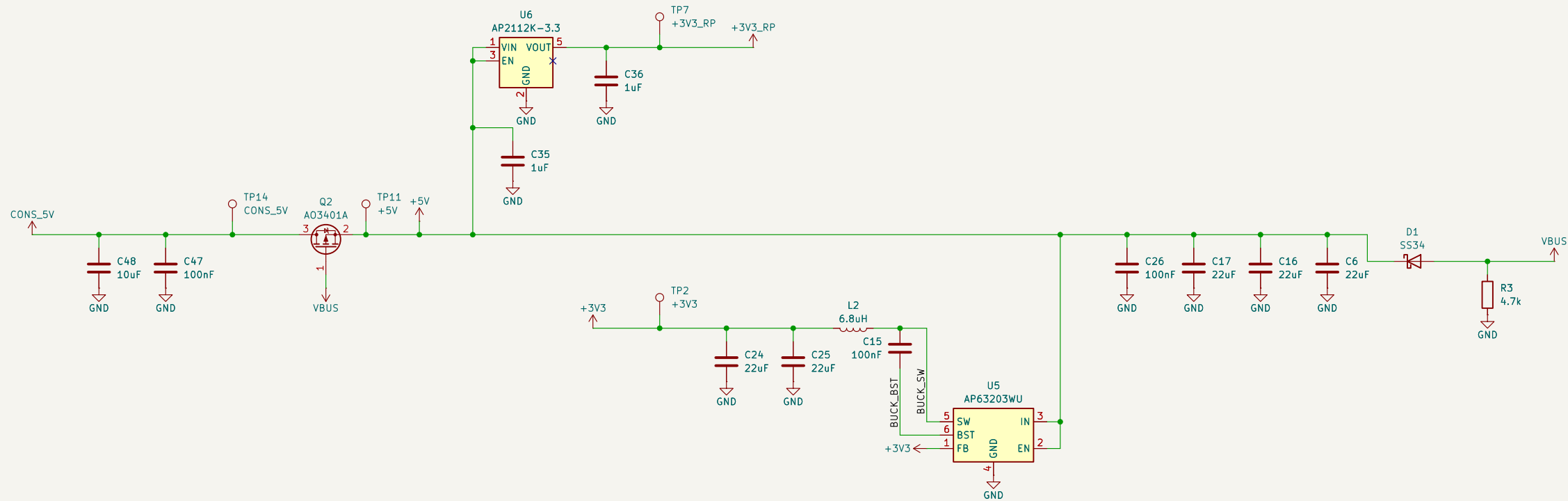
How to read this page: It is the board turned so the cartridge fingers face left (board south -> page left, north -> right, west -> top, east -> bottom) and every part sits where it is on the board. A plain wire is a connection the copper makes on one layer: the edge straight into SRAM0, which stands in the classic ROM spot over the fingers (the SMS edge is the JEDEC 32-pin memory pinout unrolled, so not one of its 21 lines crosses another), the strobes straight across to the RP2354B, the glue. The drawn buses carry what changes layers on the board: the console bus tagged by SRAM1 and the RP and the SRAM control bus (bank lines SA13-SA19, chip selects, /OE, /WE) between the RP, the glue and both SRAMs. D0-D7 reach the RP through the L008 elements beside it, /CEN and /BUSREQ end on pads. Bring-up pads: CE, OE, WE, PWR_OK (+ GND), RP2350-E9 4.7k pull-downs on GAME, LOAD, RAM_WE, MBOX, SA19 (tools/check_nets.py asserts them).

fujinet-firmware/pico/sms/sms.cartridge/sms_glue.net(), sms_glue.net(), RD, WR, CE = the console's /RD, /WR, /CE asserted; GAME, MBOX, RAM_WE, LOAD = RP GPIO30, 35, 39, 40; PWR_OK = console +5V. CEP_N = (CE & PWR_OK); WINB = A15 & A14 & A13 (the \$B500-\$BFFF load window) MBOX_N = (MBOX & A13 & A12); SLOTT2_NM = A15 & A14 & MBOX & A13 & A12; OE_ADDR = A13; SLOTT2_NM = WE_LOAD & RD & CEP & LOAD & WINB; WE_RAM = WR & CEP & RAM_WE & SLOTT2_NM; SRAM_WE = (WE_LOAD | WE_RAM); SRAM_OE = (RD & CEP & GAME & OE_ADDR); SRAM1 /CE = SA19; tools/check_glue.py evaluates this netlist against the firmware's C for every input combination.





Power tree. CONS_5V is the console's 7805 through the edge fingers: about 480 mA peak (the S3 transmitting through the buck), 180 mA average. Q2 A03401A (D = CONS_5V, S = +5V, G = VBUS) is fully on without USB, so the 74HCT glue sees VCC >= 4.5 V (a Schottky drop here would leave no margin): with USB in, its gate goes high and only its body diode conducts, so nothing back-feeds the console. D1 SS34 puts USB VBUS onto +5V. It is reverse-biased whenever the console powers the cart, and its leakage (0.5 mA at 25 C, 20 mA at 100 C) flows into VBUS, the FET gate: R3 4.7k holds VGS at -2.8 V up to 500 uA (tools/audit/spice_checks.py). +5V feeds the 5V glue and both SRAMs (cart-bus), the AP2112K LDO -> +3V3_RP (every RP2354B supply pin: it rises with the console rail, so the 5V-tolerant pads are never unpowered with 5V on them) and the AP63203 buck -> +3V3 (ESP32-S3, microSD, CP2102N). TP14, TP11, TP2, TP7 and the rp-core DVDD pad are bring-up test points.



PWR_FLAG: these rails reach their loads only through passive parts (the edge fingers, USB-C, the P-FET, the SS34, the buck's inductor).

